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Editor-in-Chief
Engineering Applications of Artificial Intelligence
Special Issue: VSI: EAAI_Agentic AI
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Dear Editor-in-Chief,

I am pleased to submit "Perceive, Plan, Act, Self-Correct: An Architectural Framework for Goal-Directed Agentic AI Systems" for consideration as a contribution to the Virtual Special Issue **VSI: EAAI_Agentic AI** in *Engineering Applications of Artificial Intelligence*.

Transfer history. This manuscript was previously submitted to *Artificial Intelligence* (AIJ), where it received a desk rejection (May 2026) on grounds of scope mismatch: the Associate Editor noted that AIJ's focus on theoretical foundations and formal proofs did not align well with the paper's empirical, engineering-oriented contributions. We believe *Engineering Applications of Artificial Intelligence* and its special issue on Agentic AI represent the natural home for this work, and no substantive changes to the manuscript have been made in response to that desk rejection.

Summary. The paper proposes the PERCEIVE–PLAN–ACT–SELF-CORRECT (PPAS) framework, a four-phase canonical loop for goal-directed agentic AI grounded in classical agent theory and extended for LLM-era architectures. We decompose agentic systems into an 8-layer technology stack mapping 15+ frameworks to a common engineering reference, validated by three empirical studies: a benchmark meta-analysis (12 agents across SWE-Bench Verified, WebArena, and GAIA), a controlled design-pattern comparison, and an inter-agent protocol evaluation (MCP, A2A, AG-UI).

Fit for EAAI and VSI: EAAI_Agentic AI. This submission is directly aligned with the special issue scope:

1. **Empirical validation of agentic architectures** — three studies producing quantitative performance data across controlled conditions and public benchmarks;
2. **Engineering-focused framework** — the 8-layer technology stack is a practical blueprint for building and evaluating real agentic systems, not a theoretical abstraction;
3. **Practical multi-agent coordination analysis** — empirical evaluation of MCP, A2A, and AG-UI protocols under controlled conditions, yielding latency and reliability measurements directly applicable to production deployments;
4. **EU AI Act governance mapping for real systems** — Article-level compliance mappings for the PPAS loop, providing a concrete engineering checklist for practitioners deploying agents under the EU regulatory framework.

Key contributions:

1. A formal four-phase agent loop unifying existing architectural patterns;
2. An 8-layer technology stack mapping 15+ frameworks to a common reference model;
3. Empirical evidence that reflective planning with human-in-the-loop approval achieves the highest task-completion rates (72.5% SWE-Bench Verified) while reducing irreversible-action failures by 73%;
4. A failure-mode taxonomy with EU AI Act and NIST AI RMF governance mappings.

Preprint. A preprint is available on engrXiv: `https://doi.org/10.31224/6738` (March 2026, accepted).

Data and code availability. All datasets, scripts, and notebooks are publicly available at: `https://github.com/HAMEEMM/ai_systems_landscape_2026/tree/main/publications/01-agentic-ai`

The manuscript is not under consideration elsewhere. I confirm compliance with the journal's submission guidelines and have disclosed the prior submission history above in the interest of full transparency. Thank you for considering this submission for the VSI: EAAI_Agentic AI special issue.

Sincerely,

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